Correlations in NYC Crime Data

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***Abstract*—NYPD Open Data provides datasets from various NYC agencies. We will analyze the Complaint Data Historic dataset by the NYPD to analyze violations, misdemeanors, and felonies to find any patterns, causes or effects of these crimes.**

I. INTRODUCTION

The NYPD Complaint Data Historic dataset includes information for crimes/complaints that are classified as violations, misdemeanors, and felonies. For each crime/complaint the data included complaint and report dates, location, time, offense descriptions, and their associated crime codes. By analyzing the data we hope to learn about each of these columns of information, their rate of occurrences, outliers, and any correlations. This information will be vital in understanding the types of crimes that happen, where they happen, and under what circumstances. This can lead to better predictions and deployment of police officers to areas and events that are most susceptible to crime. We will mostly be using Spark and Pandas to work on these goals.

II. PART I

For the first part of the project, we will analyze and generate a descriptive summary of this dataset. Simultaneously, we will also be pointing out a list of data quality issues encountered during the process.

A. Data Summary

The data summarization includes classification, verification and validation of the records in the datasets. Invalid records would essentially contain illegal values/value representation, violation of attribute dependencies, uniqueness violations (functional dependency violations) and conflicting values. The records with missing values are marked as NULL records rather than INVALID.

Following is a summary of each of the attributes present in the dataset with their descriptions -

	Column Name	Description
1	CMPLNT_NUM	Randomly generated persistent ID for each of the records.
2	CMPLNT_FR_DT	Exact date of occurrence for the reported event (Date Format)
3	CMPLNT_FR_TM	Exact time of occurrence for the reported event (Time Format)
4	CMPLNT_TO_DT	Ending date of occurrence for the reported event (Date Format)
5	CMPLNT_TO_TM	Ending time of occurrence for the reported event (Time Format)
6	RPT_DT	Date event was reported to police (Date Format)
7	KY_CD	Three digit offense classification code
8	OFNS_DESC	Description of offense corresponding with key code
9	PD_CD	Three digit internal classification code

10	PD_DESC	Description of internal classification corresponding with PD code
11	CRM_ATPT_CPTD_CD	Indicator of whether crime was successfully completed or attempted, but failed or was interrupted prematurely
12	LAW_CAT_CD	Level of offense: felony, misdemeanor, violation
13	JURIS_DESC	Jurisdiction responsible for incident. Either internal, like Police, Transit, and Housing; or external, like Correction, Port Authority, etc.
14	BORO_NM	The name of the borough in which the incident occurred
15	ADDR_PCT_CD	The precinct in which the incident occurred
16	LOC_OF_OCCUR_DESC	Specific location of occurrence in or around the premises; inside, opposite of, front of, rear of
17	PREM_TYP_DESC	Specific description of premises; grocery store, residence, street, etc.
18	PARKS_NM	Name of NYC park, playground or greenspace of occurrence, if applicable (state parks are not included)
19	HADEVELOPT	Name of NYCHA housing development of occurrence, if applicable
20	X_COORD_CD	X-coordinate for New York State Plane Coordinate System, Long Island Zone, NAD 83, units feet (FIPS 3104)
21	Y_COORD_CD	Y-coordinate for New York State Plane Coordinate System, Long Island Zone, NAD 83, units feet (FIPS 3104)
22	Latitude	Latitude coordinate for Global Coordinate System, WGS 1984, decimal degrees (EPSG 4326)
23	Longitude	Longitude coordinate for Global Coordinate System, WGS 1984, decimal degrees (EPSG 4326)
24	Lat_Lon	Coordinates of the crime scene for Global Coordinate System, WGS 1984, decimal degrees (EPSG 4326)

B. Insights

The various columns are analysed as follows -

a. Complaint_from_date (Column2)

This column represents the exact date the event occurred. These are the top 10 values for the complaints specifying dates when they occurred and the no of complaints. Overall we can see that the most complaints/crimes occur during the New year days.

01/01/2010	2343
01/01/2008	2273
01/01/2007	2259
01/01/2011	2253
01/01/2014	2211
01/01/2006	2156
08/01/2007	2053
01/01/2009	2036
01/01/2013	2030
01/01/2012	2025

b. Complaint_from_time (column3)

This column represents the exact time the crime occurred. Following top 10 values indicate that crimes are more probable after 12 pm than before 12 pm -

12:00:00	135632
15:00:00	116221
18:00:00	112102
20:00:00	106703
16:00:00	103994
17:00:00	103510
19:00:00	100290
14:00:00	92385
21:00:00	92016
22:00:00	90635

c. report_date (column6)

This column represents the date when report was reported to NYPD. The top 10 values are as follows -

5/29/2007	1903

11/01/2006	1889
10/23/2007	1854
06/21/2006	1841
06/05/2007	1829
06/20/2006	1820
07/05/2006	1803
07/29/2008	1791
05/30/2006	1785
07/31/2007	1783

d. Offense_code (column7)

This column represents the offense code and number of offenses. The top 10 values are -

341	822508
578	604141
344	521611
351	433368
109	429205
235	285801
361	283102
105	198774
107	191407
106	184124

e. Offense_description (column8)

This column represents the description for offense codes. The top 10 values are as follows -

PETIT LARCENY	822498
HARRASSMENT 2 6	04070
ASSAULT 3 & RELATED OFFENSES	521538
CRIMINAL MISCHIEF & RELATED OF	505774
GRAND LARCENY	429196
DANGEROUS DRUGS	348469
OFF. AGNST PUB ORD SENSBLTY	283065
ROBBERY	198772
BURGLARY	191406
FELONY ASSAULT	184069

As we can see 4 is not exactly equal to 5 in terms of the second column. This shows a slight error in data collection at source.

f. crime indicator (column11)

Here we have compared the count of completed vs attempted crimes.

COMPLETED	5013311
ATTEMPTED	87913

g. boro_names (column14)

This column represents the names of the borough in which the incident occurred.

BROOKLYN	1526213
MANHATTAN	1216249
BRONX	1103514
QUEENS	1011002
STATEN ISLAND	243790

h. Latitude and Longitude (Column 22 and 23)

These columns represent the coordinates of the location where the incident occurred. The analysis done checked the validity of the coordinates, whether they were a part of actual coordinates as also were they within the bounds of the coordinates of New York City

COLUMN SCRIPTS TO GENERATE BASE TYPE/SEMANTIC TYPE AND NULL/INVALID VALUES-

The column scripts are in the project folder column_details where we are generating the respective output.

The column values are given as -(number of non null entries)

CMPLNT_NUM	5101231
CMPLNT_FR_DT	5100576
CMPLNT_FR_TM	5101183
CMPLNT_TO_DT	3709753
CMPLNT_TO_TM	3713446
RPT_DT	5101231
KY_CD	5101231
OFNS_DESC	5082391
PD_CD	5096657
PD_DESC	5096657
CRM_ATPT_CPTD_CD	5101224
LAW_CAT_CD	5101231
JURIS_DESC	5101231
BORO_NM	5100768
ADDR_PCT_CD	5100841
LOC_OF_OCCUR_DESC	3974103
PREM_TYP_DESC	5067952
PARKS_NM	7599
HADEVELOPT	253205
X_COORD_CD	4913085
Y_COORD_CD	4913085
Latitude	4913085
Longitude	4913085
Lat_Lon	4913085

dtype: int64

C. Analysis

We plotted the frequency of values for each column as shown in Figure 1. As we can see most of the columns included information for each crime reported. However, there were a few

columns that had limited information. This made sense because those properties simply did not apply to all crimes. For example, the column that includes Park name is very limited because only a small fraction of all the crimes took place in NYC parks.

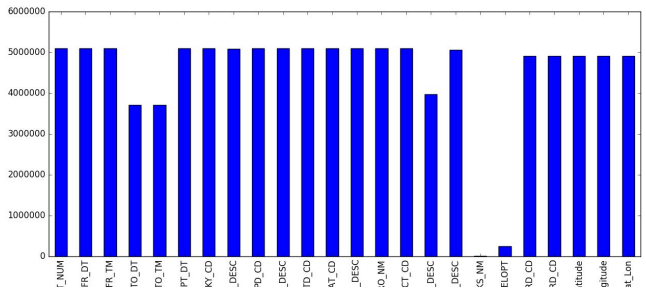

Figure 1: Frequency of values by Column

The time series graph as shown in figure 2, provided general insight of the changes in crime frequency and when they occurred. As we can see the number of crimes has reduced consistently over the years. Also crime in NYC increases significantly in the summer months and takes a plunge in the winter.

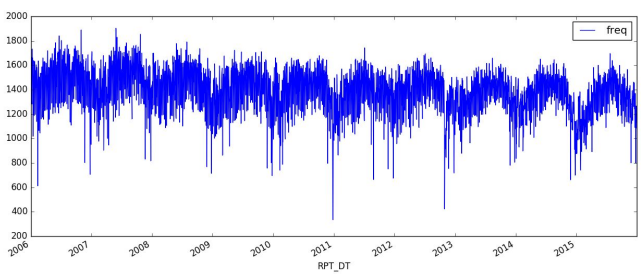

Figure 2: Time Series of Crimes

Figure 3, shows us the number of crimes that took place in each borough over the dataset time period. Brooklyn has a much higher number of crimes. This is consistent with the fact that it is the most populous borough after manhattan.

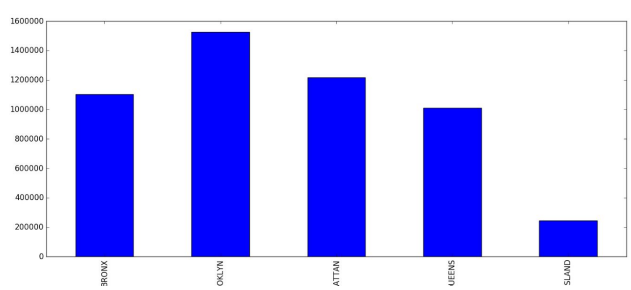

Figure 3: Frequency of Crime by Borough

III. PART II

For the second part, we looked for and explored interesting relationships between the attributes of this dataset.

Hypothesis:

1. Temperature would have a significant positive effect on crime spikes. As the temperature increases so would the total crimes committed, as the temperature would be favourable enough for outdoor crimes.
2. During holidays, we expect crime rates to increase. We speculate that crimes (indoor and misdemeanor related) would be denser during the holidays.

Temperature -

Our initial Hypothesis in Part 1 showed us that crime in NYC increases significantly in the summer months and takes a plunge in the winter.

To verify this aspect we decided to take a look at the temperature data for NY area.

Just to see if the fact is really true or if there is no relation between temperature and crime as such. We collected the data from https://www.ncdc.noaa.gov/

This Data primarily consisted of date , station name, Maximum temperature and Minimum temperature. For values which were invalid or were not recorded the dataset had -9999 as an invalid/null placeholder. The weather dataset consisted of a total 73000 rows.

After calculating the average temperature for the data using pyspark we plotted the graph for the temperature using python pandas. Before plotting the graph we did some data cleaning. We found out that out of 73000 rows 300 values for max and min temperature were inconsistent. For example, some had the value of -9999 . This could be an issue of data collection at source(ie physical sensor) or a problem in data ingestion to fill the dataset.

After filtering the 300 inconsistent values from the dataset we plotted a time series graph for the temperature over the years.

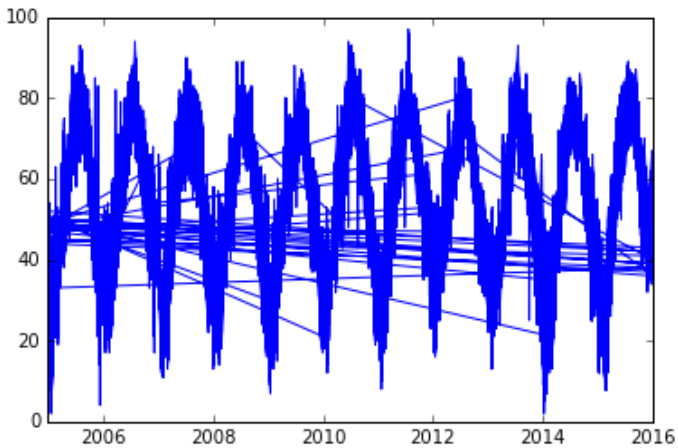

Figure 4: Temperature recorded over 2006-2016

The temperature despite of showing some minor irregularities seems to be showing an overall agreement with the crime patterns. We can hence say that crimes were more frequent during warmer seasons and less frequent over colder seasons.

Holidays using Python 'holidays' utility and plotly library-

Using the Holidays utility of Python, we tried mapping the total number of crimes with the crimes on holidays. For mapping, we used the pyplot library.

Using the pyplot library to plot these relationships, we were able to execute the tasks more efficiently than using spark or hadoop. We also found a list of illegal dates that were of the year 1015. Following are the dates -

09/16/1015

09/26/1015

10/17/1015

10/27/1015

11/25/1015

12/04/1015

Apart from these we have the following plots that represent the relationship between the crimes over a set of days.

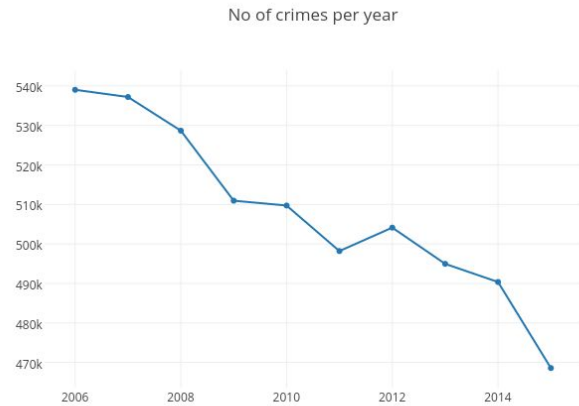

Figure 5: No of crimes recorded over 2006-2015

First we compare the number of crimes over the years. The crimes as we can see decreased gradually starting from 2006 till 2015 with an unusual spike in the year 2012.

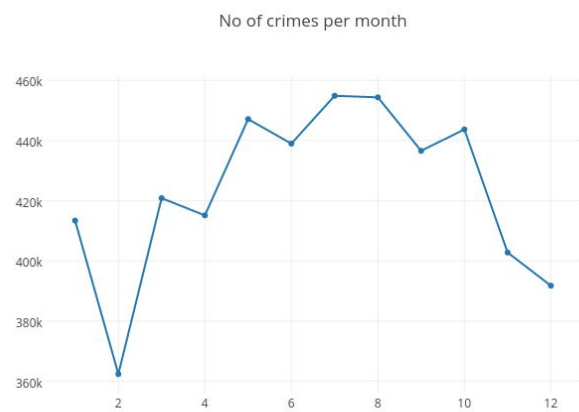

Figure 6: No of crimes recorded in each month over 2006-2015

The total number of crimes each month has been plotted in the above graph. The cumulative number of crimes in February being the least and that in July and August to be the most. One inference from this observation is the fact that February has 28/29 days in a year which is always lesser than the number of days in the rest of the months. Also July and August are the only consecutive months during the year to have 31 days, may be that might be a factor in the increased number of crimes apart from the fact that they are in general also months with higher temperature averages.

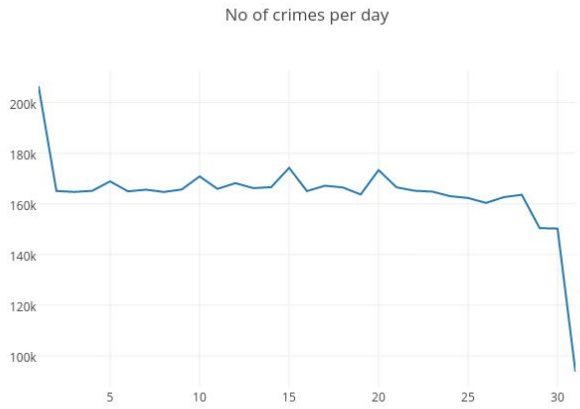

Figure 7: No of crimes recorded each day of the month over 2006-2015

The total number of crimes for each day of the month for years 2006-2015 have been plotted in the above graph. It is obvious that the number of crimes for the date 31 is less in comparison to other days, the reason being, the date 31 comes in only for 7 months out of the 12. Also in general the deviation in the number of crimes for the dates 2-29 is less than 10%. The number of crimes for the date 30 is slightly lesser, the reason being, it not being present for the month of February. However, an outlying revelation out of this graph is the number of crimes on the very 1st day of each month, which is significantly higher than the general population.

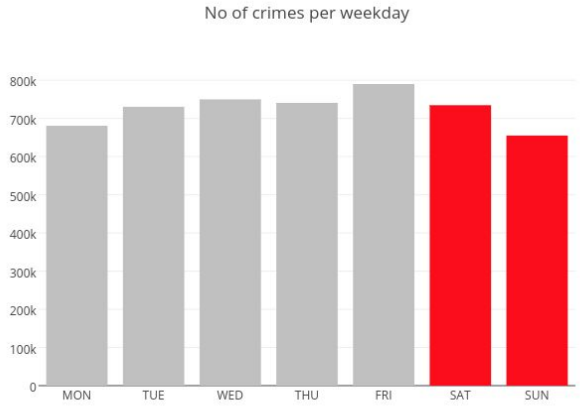

Figure 8: No of crimes recorded over each day of the week over 2006-2015

The above relationship is the mapping of the number of crimes for each day of the week. This relationship is by far the most sensible and informative in comparison to the rest. There is no particular skewing of data in this comparison, like the ones we came across in the plotting of number of crimes for each month and date of the month. The number of crimes on the weekends (Saturday and Sunday) turns out to be slightly lesser than the number of crimes on weekdays. The number of crimes during the middle of the week, i.e. Wednesday, Thursday and Friday are in higher than the number on Mondays and weekends. There could be several reasons for such a result, ranging from weekends being the time when people are less frustrated and Monday being the start of the week, people normally have a tendency to have a fresh start on this day. On the other hand, Friday has the highest number of crimes, one of the reasons could be the fact that the behaviour of people in general deteriorates as they reach the end of the week and anticipate a weekend.

The correlation coefficient between the number of crimes during the weekdays in comparison to the crimes during the weekends is 0.0076.

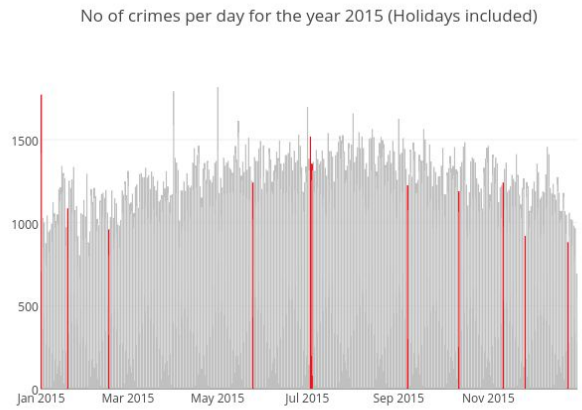

Figure 9: No of crimes per day recorded over 2006-2015

The above graph plots the number of crimes each day during the entire year of 2015. The plots in red color are the crimes committed on holidays. Based on the graph, January 1st or New Years is one of the days with the highest number of crimes during the entire year with similar spikes only twice in the year, once in April and the other in May, both of which weren't federal holidays. The graph does not reveal any kind of astonishing detail but in general, the number of crimes on a holiday seem to be lesser in comparison to the number of crimes during the surrounding days of the holidays. This could be due to several reasons like people in general do not commit crimes on holidays or may authorities are slightly more forgiving on these days to actually file any kind of complaint on that day.

The correlation coefficient between the number of crimes during the normal days in comparison to the crimes during the holidays is 0.00011.

Holidays mean analysis -

Holidays tend to be disruptions to the average employee's work week. We mapped several instances of the federal holidays against their respective months average daily crimes to observe any possible correlations between these holidays and the crimes committed. Our holiday dataset included ten federal holidays:

- New Year's Day
- Martin Luther King Jr. Day
- Presidents Day
- Memorial Day
- Independence Day
- Labor Day
- Columbus Day
- Veterans Day
- Thanksgiving Day
- Christmas Day

Our holiday dataset had to be adjusted in order to make the date format analogous to the date format used in the crimes dataset. New Years, Independence, and Christmas Days are all holidays that are celebrated on a specific day every year. However, if these holidays fall on a weekend then the federal holiday is set for either the friday before or the monday after. Regardless, of which day the federal holiday is set for, the actual holiday is celebrated by Americans on its original day. Our dataset included the federal dates and needed to be adjusted to represent the dates in which the holidays were actually celebrated.

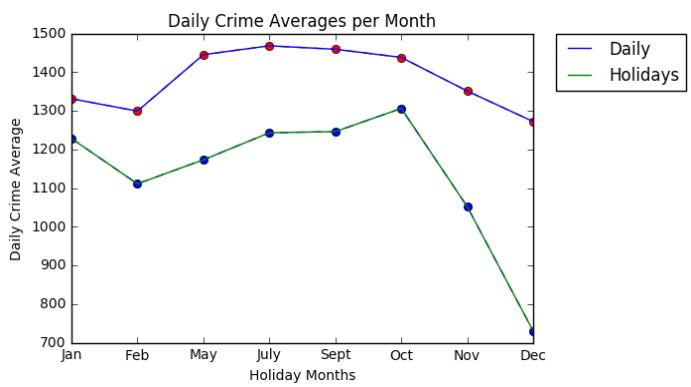

Figure 10: Temperature recorded over 2006-2016

The graph above plots the average daily crimes in a particular month and the total daily crimes that took place during the holidays of respective month. Only months that contain federal holidays are represented here. As we can see contrary to our hypothesis, total crimes during the holidays took a plunge. There may be several reasons for this. In particular, the NYPD has more officers patrolling during holidays, which drastically deters many potential crimes especially during Christmas.

We wanted to look deeper into the data to see if there were any types of crimes that rejected this trend. Below are 7 bar graphs (one for each holiday month) that plot crimes per category for both daily averages and holidays.

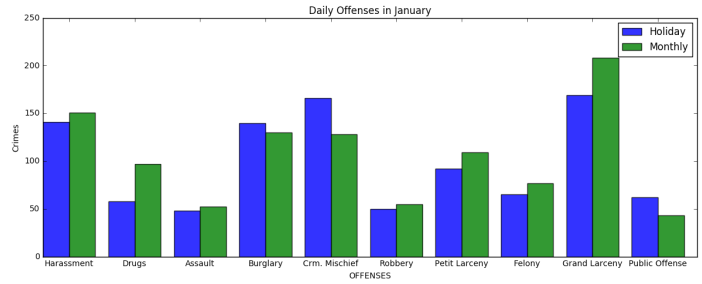

Figure 11: Avg no of crimes in January

As we can see for the month of January, Burglary, Criminal Mischief, and Public Offenses are higher than average. Everything else is below average.

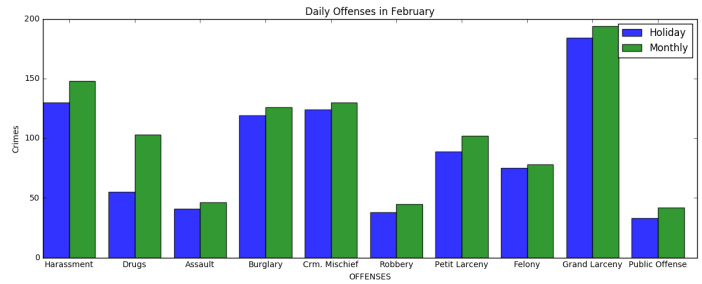

Figure 12: Avg no of crimes in February

For the month of February, all types of crimes are committed less frequently during the holiday than on average.

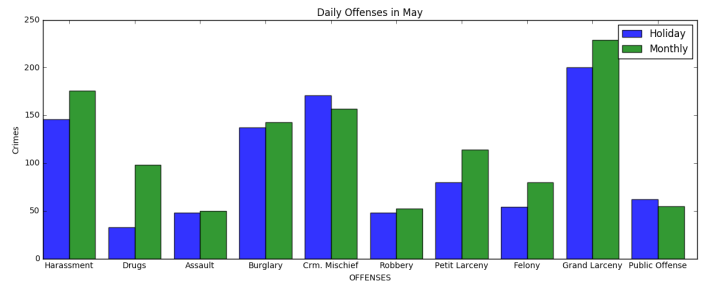

Figure 13: Avg no of crimes in May

For the month of May, Criminal Mischief, and Public Offenses are higher than average. Everything else is below average.

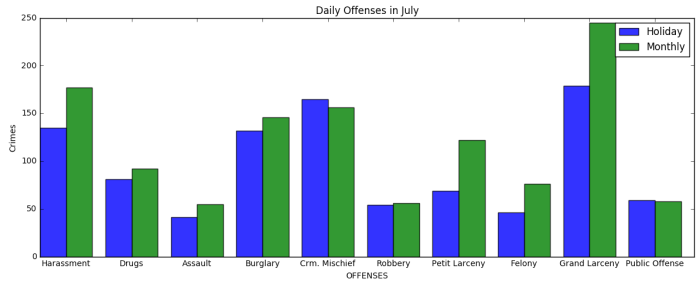

Figure 14: Avg no of crimes in July

For the month of July, Criminal Mischief, and Public Offenses are higher than average. Everything else is below average.

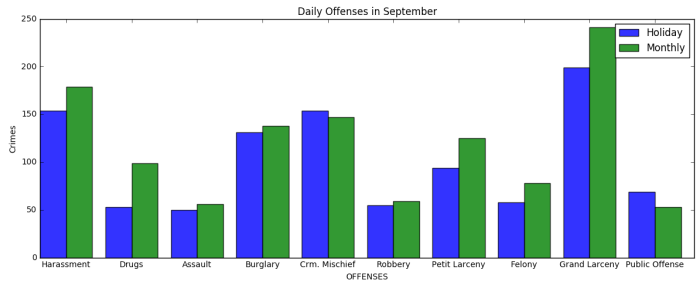

Figure 15: Avg no of crimes in September

For the month of September, Criminal Mischief, and Public Offenses are higher than average. Everything else is below average.

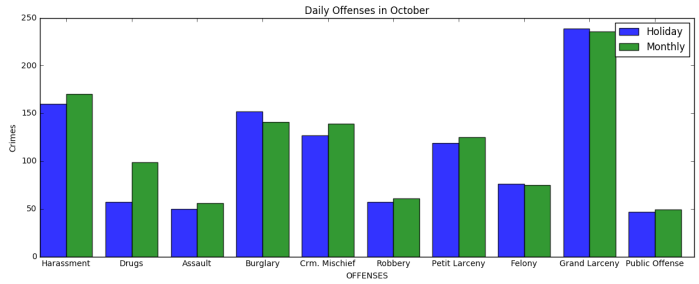

Figure 16: Avg no of crimes in October

For the month of October, Burglary, and Grand Larceny are higher than average. Everything else is below average.

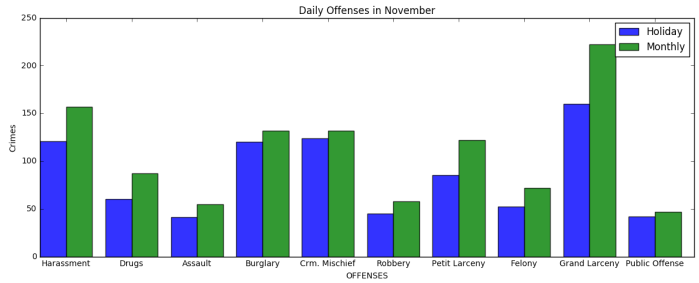

Figure 17: Avg no of crimes in November

For the month of November, all types of crimes are committed less frequently during the holiday than on average.

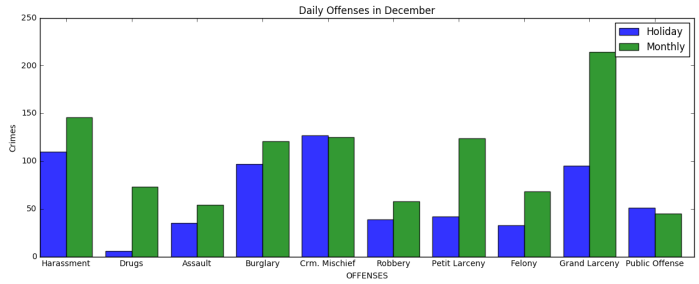

Figure 18: Avg no of crimes in December

For the month of December, Criminal Mischief, and Public Offenses are slightly higher than average. Everything else is below average. Grand Larceny, petit larceny, and drugs are significantly below average.

Overall, the data tells us that crimes committed decreases during holidays. However, public offenses and criminal mischiefs take a very slight uptick during the holidays. Perhaps this is because there are more people out and about instead of at work. We can imagine that a similar correlation can be found for criminal mischief and public offenses committed in the weekends over the weekdays.

IV. INDIVIDUAL CONTRIBUTIONS

Karan -

Contributed to initial aggregation of columns in the summary folder of the project.Derived key initial insights. Wrote the Spark scripts to get summary for -

i. Complaint_from_date
ii. Complaint_from_time
iii. report_date
iv. Offense_code
v. Offense_description
vi. crime indicator
vii. boro_names

Contributed to the time series analysis. In the analysis folder of the project - crime_analysis.py. Generated the plots using the code. Wrote code using python pandas for the functions generating these graphs -

viii. Frequency of values by Column
ix. Time Series of Crimes
x. Frequency of Crime by Borough

Retrieved invalid and null values during the pre aggregation phase of crime analysis

Collected temperature dataset . Wrote the spark script temp_summary.py to get the dates and the average temperature.

Resolved data quality issues for 300 values in that same script. This was our First Hypothesis.

Analysed the temperature dataset .Wrote temperature_analysis.ipynb to provide an overall aggregation of temperature from 2006-2016

Contributed to retrieving the base type ,semantic type and the null /invalid values for two columns

xi. Report Date
xii. Complaint time from
xiii. Borough Name
xiv. Key Code
xv. Latitude
xvi. Longitude

Contributed to create report for Insights and summary.

Rohul -

a. Contributed to initial aggregation of columns in the summary folder of the project
b. Contributed to the time series analysis. In the analysis folder of the project - crime_analysis.py
c. Initial draft of the report
d. Contributed to retrieving the base type ,semantic type and the null /invalid values for two columns
   xvii. Complaint Number
   xviii. Borough Name
   xix. Key Code
   xx. Offense Description
   xxi. PD Code
   xxii. PD Description
   xxiii. Crime Completed/Attempted
   xxiv. Level of Offense.
e. Wrote code to correlate Holiday datasets, analyzed graphs, and discussed related findings.

Divyesh -

a. Contributed to the initial aggregation of columns in the summary folder of the project
b. Contributed to the time series analysis. In the analysis folder of the project - crime_analysis.py. Generated the plots using the code
c. Developed and coded the time series analysis for the total number of crimes over various aspects using Plotly library
d. Drafting of the report
e. Retrieved the base type, semantic type and the null /invalid values for five columns
   i. X_COORD_CD
   ii. Y_COORD_CD
   iii. Latitude
   iv. Longitude
   v. LatLong
f. Used plotly to correlate and visualize the holiday dataset with the crime dataset, and discussed the related findings for better understanding

Our tasks were divided between finding data summary and data quality issues. For the first part we split up the columns and took on looking at the data type, semantic type, and label and looking for any issues with each value. For part II, we split up the work more distinctly based on the hypothesis and the analysis we took up to be conducted over the dataset.

V. CONCLUSION

Overall for part 1, we can see that most of the data is clean and easy to work with. There are a few issues with data quality including inconsistency between offense description and code. The plot of the data gives us insight into the crimes that take place. We can see that Brooklyn is the most crime ridden of the five boroughs, which makes sense because Brooklyn is the most populated borough, and that crime takes an uptick during the summer months.

In conclusion, as a part of the analysis, we came across findings that either supported or debunked our initial hypothesis. The increase in temperature did affect the number of crimes positively. Warmer temperatures recorded more number of crimes. On the other hand, our hypothesis of more number of crimes on holidays wasn't entirely satisfied. Crimes were astonishingly high during certain holidays but for most were comparatively low in comparison to the normal working days. The number of crimes over the years seemed to have decreased. Also, the number of crimes would spike in the beginning of each month and would be lesser for weekends than weekdays.

Overall, the work helped us understand the dataset of the complaints recorded and, correlating it with other datasets helped us visualize and understand the factors that may have caused certain trends in the dataset.

VI. CODE

https://github.com/divsurana/csgy9223bigdata

VII. REFERENCES

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